Year 10 Foundation

REVISION PACK

Shape and Space 1

Name: _____

Class: _____

Shape and Space 1 · Angles (Lessons 5.1, 5.2)

Worked example: the exterior angle of a triangle

BCD is a straight line. The two interior opposite angles are 48° and 65°. Work out the size of angle x. Give a reason for your answer.

Step 1	Exterior = two opposite angles	$x = 48 + 65$
Step 2	Answer	$x = 113^\circ$
Step 3	Reason	exterior angle of a triangle

Practice: the exterior angle of a triangle

1.1

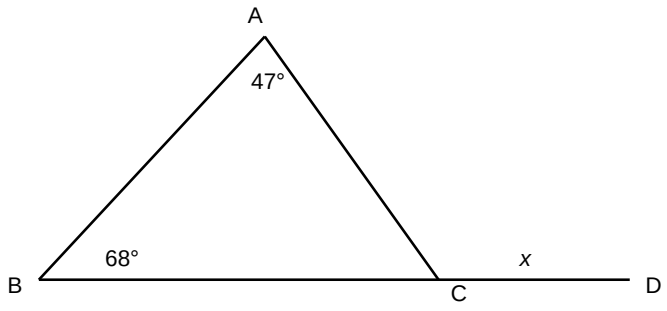


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

1.2

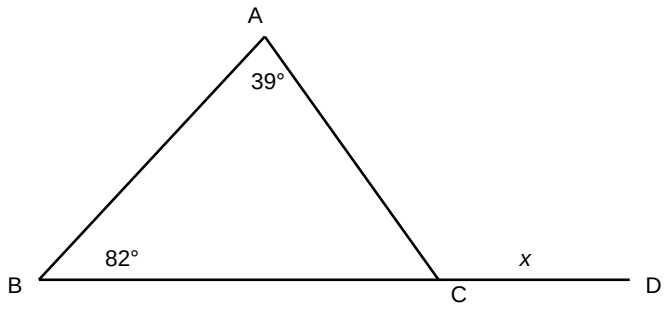


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

1.3

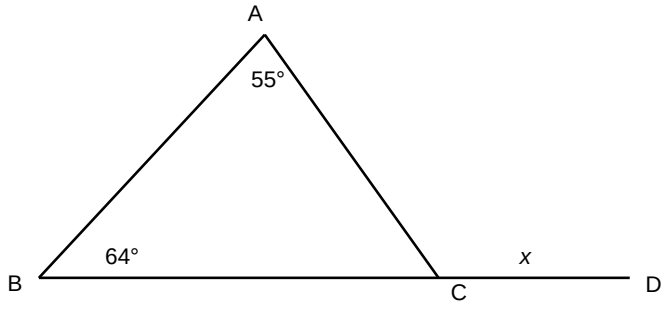


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

Practice: the exterior angle of a triangle (continued)

1.4

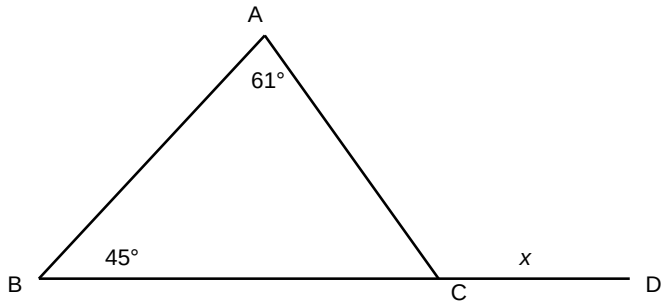


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

1.5

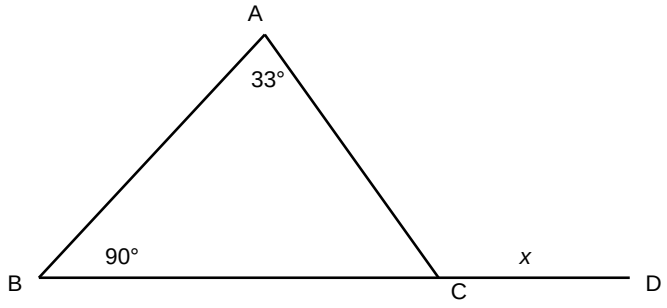


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

1.6

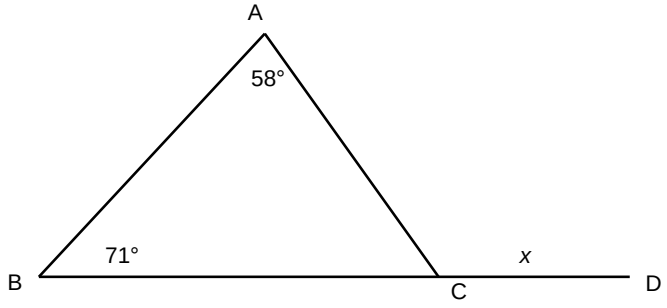


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

Shape and Space 1 · Angles (Lessons 5.1, 5.2)

Worked example: angles in a quadrilateral

Three angles of a quadrilateral are 95°, 80° and 110°. Work out the size of the fourth angle, x.

Step 1	Angles add to 360°	$95 + 80 + 110 = 285$
Step 2	Take it from 360°	$360 - 285 = 75$
Step 3	Answer	$x = 75^\circ$

Practice: angles in a quadrilateral

1.7

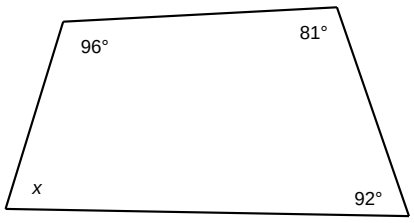


Diagram **NOT** accurately drawn

Work out the size of angle x.

1.8

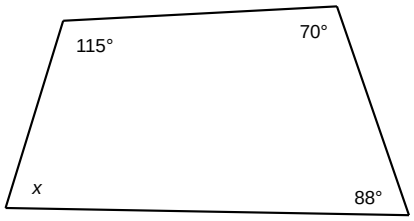


Diagram **NOT** accurately drawn

Work out the size of angle x.

1.9

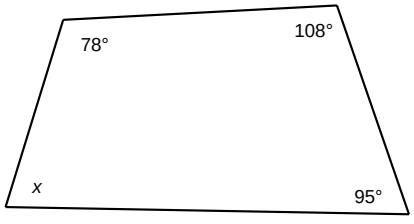


Diagram **NOT** accurately drawn

Work out the size of angle x.

Practice: angles in a quadrilateral (continued)

1.10

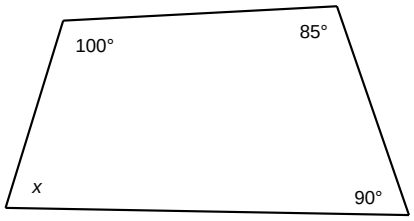


Diagram **NOT** accurately drawn

Work out the size of angle x.

1.11

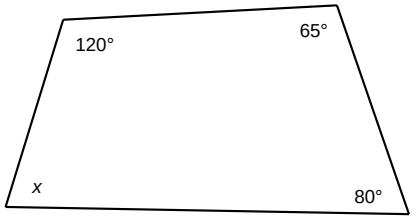


Diagram **NOT** accurately drawn

Work out the size of angle x.

1.12

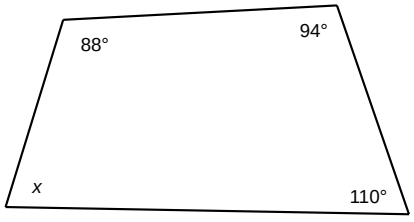


Diagram **NOT** accurately drawn

Work out the size of angle x.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Worked example: finding the hypotenuse

The two shorter sides of a right-angled triangle are 6 cm and 8 cm. Work out the length of the hypotenuse.

Step 1	Square and add (Pythagoras)	$x^2 = 6^2 + 8^2 = 100$
Step 2	Square root	$x = \sqrt{100}$
Step 3	Answer	10 cm

Practice: finding the hypotenuse

2.1

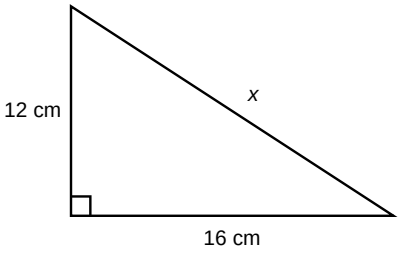


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

2.2

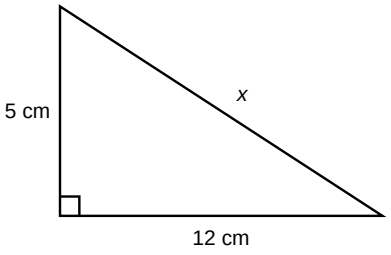


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

2.3

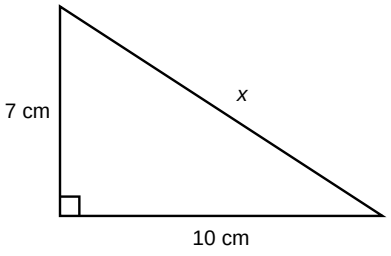


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.
Give your answer correct to 3 significant figures.

Practice: finding the hypotenuse (continued)

2.4

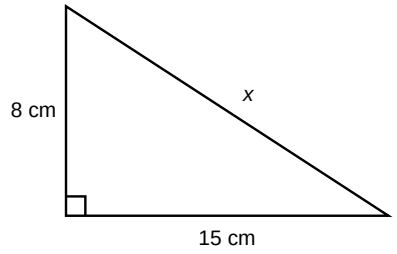


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

2.5

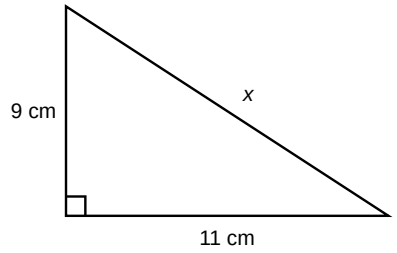


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x. Give your answer correct to 3 significant figures.

2.6

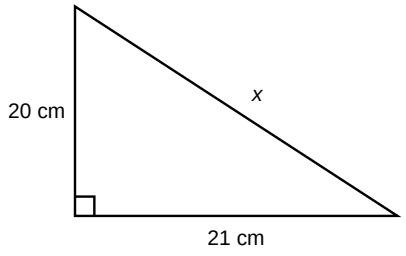


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Worked example: compound shapes

A shape is made from a rectangle 10 cm by 4 cm and a triangle. The triangle has base 10 cm and perpendicular height 3 cm. Work out the total area of the shape.

Step 1	Rectangle	$10 \times 4 = 40$
Step 2	Triangle	$\frac{1}{2} \times 10 \times 3 = 15$
Step 3	Add the two areas	$40 + 15 = 55$
Step 4	Answer	$55\ \text{cm}^2$

Practice: compound shapes

2.7

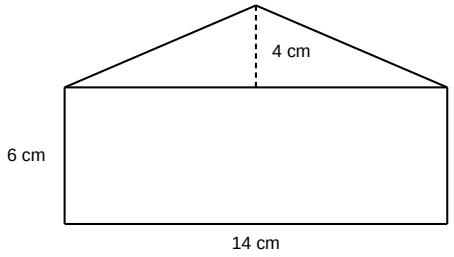


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 4 cm.
Work out the total area of the shape.

2.8

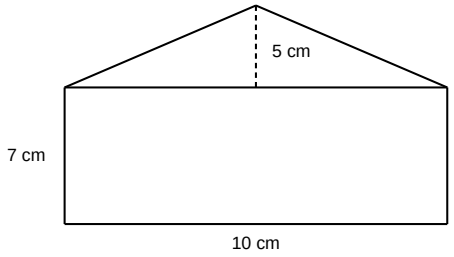


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 5 cm.
Work out the total area of the shape.

2.9

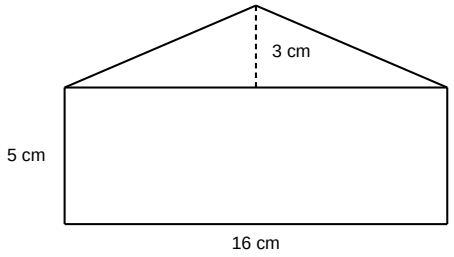


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 3 cm.
Work out the total area of the shape.

Practice: compound shapes (continued)

2.10

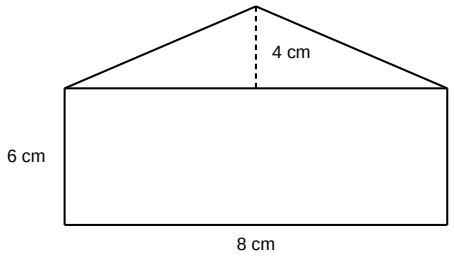


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 4 cm.
Work out the total area of the shape.

2.11

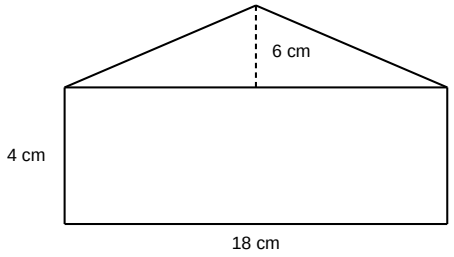


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 6 cm.
Work out the total area of the shape.

2.12

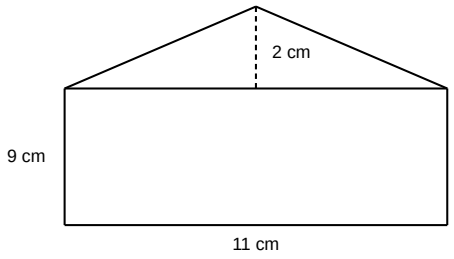


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 2 cm.
Work out the total area of the shape.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Worked example: trapezium problems

A lawn is a trapezium with parallel sides 8 m and 12 m and height 5 m. One box of grass seed covers 15 m². How many boxes are needed to cover the lawn?

Step 1	Area	$\frac{1}{2}(8+12)\times 5 = 50\ \text{m}^2$
Step 2	Area ÷ area one box covers	$50 \div 15 = 3.33\ldots$
Step 3	Round up: can't buy part	4 boxes

Practice: trapezium problems

2.13

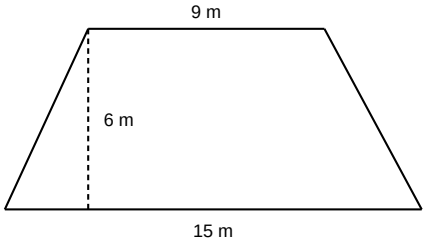


Diagram **NOT** accurately drawn

Ahmad's lawn is a trapezium.
One box of grass seed covers 20 m²
How many boxes must Ahmad buy to cover the lawn?

2.14

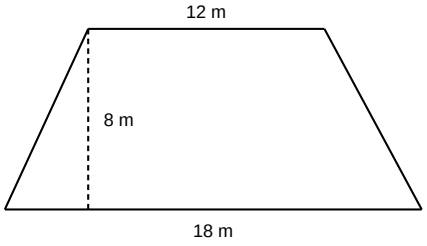


Diagram **NOT** accurately drawn

Noor's lawn is a trapezium.
One box of grass seed covers 25 m²
How many boxes must Noor buy to cover the lawn?

2.15

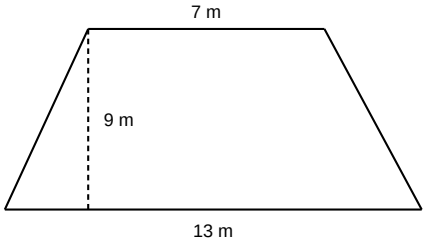


Diagram **NOT** accurately drawn

Faisal's lawn is a trapezium.
One box of grass seed covers 25 m²
How many boxes must Faisal buy to cover the lawn?

Practice: trapezium problems (continued)

2.16

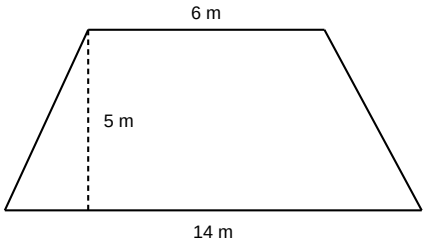


Diagram **NOT** accurately drawn

Sara's lawn is a trapezium.
One box of grass seed covers 15 m²
How many boxes must Sara buy to cover the lawn?

2.17

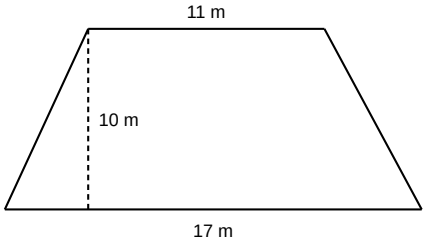


Diagram **NOT** accurately drawn

Yousef's lawn is a trapezium.
One box of grass seed covers 30 m²
How many boxes must Yousef buy to cover the lawn?

2.18

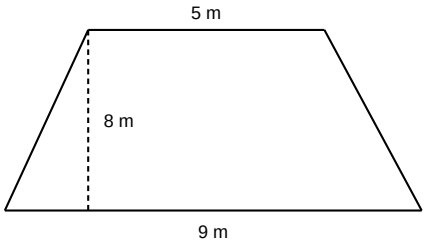


Diagram **NOT** accurately drawn

Maha's lawn is a trapezium.
One box of grass seed covers 10 m²
How many boxes must Maha buy to cover the lawn?

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Worked example: area of a circle

The diameter of a circle is 10 cm. Work out the area of the circle. Give your answer correct to 3 significant figures.

Step 1	Radius = diameter ÷ 2	$10 \div 2 = 5$
Step 2	πr^2	$\pi \times 5^2 = 78.5398\ldots$
Step 3	Answer	$78.5\ \text{cm}^2$

Practice: area of a circle

2.19

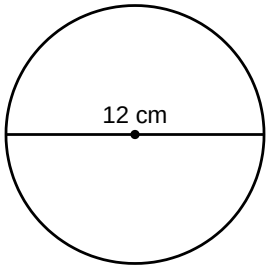


Diagram **NOT** accurately drawn

The diameter of the circle is 12 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

2.20

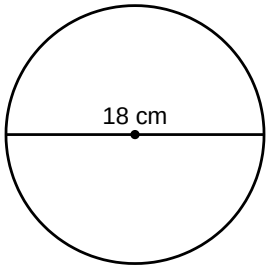


Diagram **NOT** accurately drawn

The diameter of the circle is 18 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

2.21

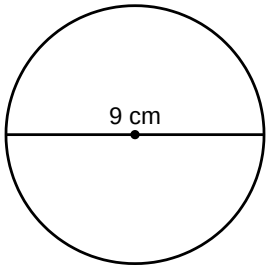


Diagram **NOT** accurately drawn

The diameter of the circle is 9 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

Practice: area of a circle (continued)

2.22

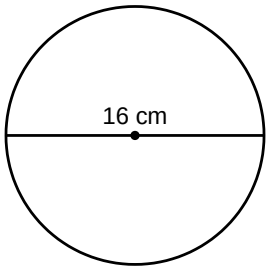


Diagram **NOT** accurately drawn

The diameter of the circle is 16 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

2.23

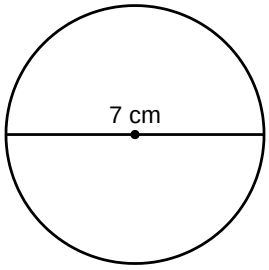


Diagram **NOT** accurately drawn

The diameter of the circle is 7 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

2.24

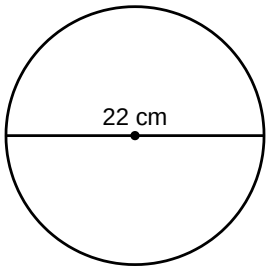


Diagram **NOT** accurately drawn

The diameter of the circle is 22 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

Worked example: finding a shorter side

A right-angled triangle has a hypotenuse of length 13 cm. One of the shorter sides is 5 cm. Work out the length of the other side.

Step 1	Hypotenuse squared minus the side squared	$13^2 - 5^2 = 169 - 25 = 144$
Step 2	Square root	$\sqrt{144} = 12$
Step 3	Answer	12 cm

Practice: finding a shorter side

3.1 A right-angled triangle has a hypotenuse of length 15 cm. One of the shorter sides is 9 cm long. Work out the length of the other side. Give your answer correct to 3 significant figures where necessary.

3.2 A right-angled triangle has a hypotenuse of length 17 cm. One of the shorter sides is 8 cm long. Work out the length of the other side. Give your answer correct to 3 significant figures where necessary.

3.3 A right-angled triangle has a hypotenuse of length 12 cm. One of the shorter sides is 7 cm long. Work out the length of the other side. Give your answer correct to 3 significant figures where necessary.

3.4 A right-angled triangle has a hypotenuse of length 9 cm. One of the shorter sides is 4 cm long. Work out the length of the other side. Give your answer correct to 3 significant figures where necessary.

3.5 A right-angled triangle has a hypotenuse of length 20 cm. One of the shorter sides is 11 cm long. Work out the length of the other side. Give your answer correct to 3 significant figures where necessary.

3.6 A right-angled triangle has a hypotenuse of length 8.5 cm. One of the shorter sides is 4 cm long. Work out the length of the other side. Give your answer correct to 3 significant figures where necessary.

Worked example: circumference

A circle has a radius of 4.5 cm. Work out the circumference. Give your answer correct to 1 decimal place.

Step 1	Use $C = 2\pi r$	$2 \times \pi \times 4.5$
Step 2	Calculator	$28.274\ldots$
Step 3	Answer	28.3 cm

Practice: circumference

3.7 A circle has a radius of 3 cm. Work out the circumference. Give your answer correct to 1 decimal place.

3.8 A circle has a diameter of 12 cm. Work out the circumference. Give your answer correct to 1 decimal place.

3.9 A circle has a radius of 7.5 cm. Work out the circumference. Give your answer correct to 1 decimal place.

3.10 A bicycle wheel has a diameter of 25 cm. Work out the circumference of the wheel. Give your answer correct to 1 decimal place.

3.11 A circular pond has a radius of 10 m. Work out the circumference of the pond. Give your answer correct to 1 decimal place.

3.12 A plate has a diameter of 8.4 cm. Work out the circumference of the plate. Give your answer correct to 1 decimal place.

Shape and Space 1 · Angles (Lesson 5.1)

Worked example: angles in an isosceles triangle

Triangle ABC is isosceles with AB = AC. Angle BAC = 40°. Work out the size of angle ABC.

Step 1	Angles in a triangle add to 180°	$180 - 40 = 140$
Step 2	The two base angles are equal	$140 \div 2 = 70$
Step 3	Answer	70°

Practice: angles in an isosceles triangle

3.13 Triangle ABC is isosceles with AB = AC. Angle BAC = 50°. Work out the size of angle ABC.

3.14 Triangle PQR is isosceles with PQ = PR. Angle QPR = 100°. Work out the size of angle PQR.

3.15 Triangle DEF is isosceles with DE = DF. Angle EDF = 36°. Work out the size of angle DEF.

3.16 Triangle ABC is isosceles with AB = AC. Angle ABC = 55°. Work out the size of angle BAC.

3.17 Triangle XYZ is isosceles with XY = XZ. Angle XYZ = 72°. Work out the size of angle YXZ.

3.18 Triangle LMN is isosceles with LM = LN. Angle MLN = 124°. Work out the size of angle LMN.